Quasi-invariance of the Φ^4_3 measure under smooth shifts

Abstract

Let μ denote the Φ^4_3 (Euclidean) measure on the three-dimensional unit torus $\mathbb{T}^3$. We show that for every smooth (nonzero) function ψ on $\mathbb{T}^3$, the translate $T^*_\psi\mu$ is equivalent to μ (mutual absolute continuity). The proof proceeds by expressing μ as a strictly positive Gibbs density with respect to the massive Gaussian free field and invoking the Cameron–Martin theorem. An explicit Radon–Nikodym derivative is also recorded.

1 Setup and a convenient definition of the Φ^4_3 measure

Let $\mathbb{T}^3 := (\mathbb{R}/\mathbb{Z})^3$ be the unit 3-torus and let $\mathcal{D}'(\mathbb{T}^3)$ be the space of Schwartz distributions on $\mathbb{T}^3$. For $\psi \in C^\infty(\mathbb{T}^3)$ we denote by

$$T_\psi : \mathcal{D}'(\mathbb{T}^3) \to \mathcal{D}'(\mathbb{T}^3), \qquad T_\psi(u) = u + \psi$$

the translation (shift) map.

Fix a mass parameter $m > 0$ and set

$$A := m^2 - \Delta \quad \text{on } \mathbb{T}^3,$$

so that A is a strictly positive self-adjoint operator on $L^2(\mathbb{T}^3)$. Let ν be the centered Gaussian (massive free field) probability measure on $\mathcal{D}'(\mathbb{T}^3)$ with covariance operator $C := A^{-1}$. Equivalently, if u is a random distribution with law ν then for every test function $\varphi \in C^\infty(\mathbb{T}^3)$,

$$\langle u, \varphi\rangle \text{ is centered Gaussian with variance } \mathbb{E}_\nu[\langle u, \varphi\rangle^2] = \langle \varphi, C\varphi\rangle_{L^2}.$$

(Here and below $\langle u, \varphi\rangle$ denotes the distribution–test-function pairing.)

Wick powers with respect to ν can be defined in the usual way via approximation by mollification and subtraction of diverging constants, or equivalently via the homogeneous Wiener chaos decomposition. In particular, for each integer $k \geq 1$ one obtains a random distribution $:u^k:\in \mathcal{D}'(\mathbb{T}^3)$ such that for every $\varphi \in C^\infty(\mathbb{T}^3)$ the pairing $\langle :u^k:, \varphi\rangle$ is a finite random variable (indeed in $L^p(\nu)$ for every $p < \infty$). We will only need $k = 2, 3, 4$.

Fix a coupling constant $\lambda > 0$ and a (renormalized) quadratic coefficient $\rho \in \mathbb{R}$. Define the renormalized interaction functional

$$\mathcal{V}(u) := \int_{\mathbb{T}^3} \left(\frac{\lambda}{4} :u(x)^4: + \frac{\rho}{2} :u(x)^2: \right) dx, \tag{1}$$

where the integral is understood as the pairing of the random distribution $\frac{\lambda}{4} :u^4: + \frac{\rho}{2} :u^2:$ with the constant test function 1. This quantity is finite ν-almost surely by the above remarks.

The (finite volume) Φ^4_3 measure μ is then given by the Gibbs perturbation of ν

$$\mu(du) := Z^{-1} \exp\bigl(-\mathcal{V}(u)\bigr)\, \nu(du), \qquad Z := \int \exp\bigl(-\mathcal{V}(u)\bigr)\, \nu(du) \in (0,\infty). \tag{2}$$

The finiteness and positivity of Z for $\lambda > 0$ is the well-known integrability statement in constructive field theory for the (super-renormalizable) $P(\varphi)_3$ models in finite volume; we take it as part of the standard meaning of "Φ^4_3 measure" on $\mathbb{T}^3$.

Remark 1. *Different renormalization conventions merely alter ρ and multiply the density by an overall (finite) constant absorbed into Z; none of this affects the quasi-invariance statement under smooth shifts.*

2 Two general lemmas

Lemma 1 (Equivalence to the reference Gaussian measure)**.** *Let ν be a probability measure on a measurable space $(E, \mathcal{E})$ and let $\mu(du) = Z^{-1} e^{-W(u)} \nu(du)$ where $Z \in (0,\infty)$ and $W : E \to \mathbb{R}$ is measurable and finite ν-a.s. Then μ and ν are equivalent (mutually absolutely continuous).*

Proof. Clearly $\mu \ll \nu$. Conversely, if $\mu(A) = 0$ for some $A \in \mathcal{E}$ then

$$0 = \mu(A) = Z^{-1} \int_A e^{-W(u)}\, \nu(du).$$

Since $e^{-W(u)} > 0$ for ν-almost every u, the integral can vanish only if $\nu(A) = 0$. Hence $\nu \ll \mu$. □

Lemma 2 (Equivalence is preserved under measurable maps)**.** *Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces, $T : E \to F$ measurable, and let μ, ν be equivalent measures on E. Then $T_*\mu$ and $T_*\nu$ are equivalent on F.*

Proof. For $B \in \mathcal{F}$, $(T_*\mu)(B) = \mu(T^{-1}B)$ and $(T_*\nu)(B) = \nu(T^{-1}B)$. Since μ and ν have the same null sets, $\mu(T^{-1}B) = 0$ iff $\nu(T^{-1}B) = 0$, hence $T_*\mu$ and $T_*\nu$ have the same null sets. □

3 Cameron–Martin theorem for the free field

The Cameron–Martin space of the centered Gaussian measure ν with covariance $C = A^{-1}$ is the Sobolev space $H^1(\mathbb{T}^3)$ equipped with the inner product

$$\langle h_1, h_2 \rangle_{H^1_A} := \langle A^{1/2} h_1, A^{1/2} h_2 \rangle_{L^2} = \langle h_1, A h_2 \rangle_{L^2}, \qquad \|h\|^2_{H^1_A} = \langle h, Ah \rangle_{L^2}.$$

(For $m > 0$, this norm is equivalent to the standard H^1 norm.)

Proposition 1 (Cameron–Martin for ν)**.** *If $\psi \in H^1(\mathbb{T}^3)$, then $T^*_\psi \nu$ is equivalent to ν and*

$$\frac{d(T^*_\psi \nu)}{d\nu}(u) = \exp\left(\langle u, A\psi \rangle - \frac{1}{2} \langle \psi, A\psi \rangle_{L^2} \right). \tag{3}$$

In particular, this holds for every $\psi \in C^\infty(\mathbb{T}^3)$.

Proof. This is the standard Cameron–Martin theorem for centered Gaussian measures. Since $\psi \in C^\infty$ implies $A\psi \in C^\infty$, the random variable $\langle u, A\psi \rangle$ is well-defined as a Gaussian linear functional of u under ν, and (**??**) is the usual finite-dimensional formula extended to the infinite-dimensional Gaussian setting. □

4 Main result: equivalence of μ and its smooth translate

Theorem 1. *Let μ be the Φ^4_3 measure on $\mathcal{D}'(\mathbb{T}^3)$ defined by (**??**) and let $\psi \in C^\infty(\mathbb{T}^3)$ be nonzero (or arbitrary). Then μ and $T^*_\psi\mu$ are equivalent measures. Moreover, $T^*_\psi\mu \ll \mu$ with Radon–Nikodym derivative*

$$\frac{d(T^*_\psi\mu)}{d\mu}(u) = \exp\left(\langle u, A\psi\rangle - \frac{1}{2}\langle \psi, A\psi\rangle_{L^2} + \mathcal{V}(u) - \mathcal{V}(u-\psi)\right), \tag{4}$$

*where $\mathcal{V}$ is given by (**??**). In particular, this derivative is strictly positive and finite μ-almost surely, and its reciprocal is obtained by replacing ψ with $-\psi$.*

Proof. **Step 1: μ is equivalent to ν.** By construction (**??**), the density $Z^{-1}e^{-\mathcal{V}(u)}$ is finite and strictly positive ν-a.s. Therefore Lemma **??** yields $\mu \sim \nu$.

Step 2: the free field is quasi-invariant under T_ψ. By Proposition **??**, since $\psi \in C^\infty(\mathbb{T}^3) \subset H^1(\mathbb{T}^3)$, we have $T^*_\psi\nu \sim \nu$.

Step 3: chain of equivalences. Using Lemma **??** and Step 1 we obtain

$$T^*_\psi\mu \sim T^*_\psi\nu.$$

Using Step 2 gives $T^*_\psi\nu \sim \nu$, and using Step 1 again gives $\nu \sim \mu$. By transitivity of equivalence, $T^*_\psi\mu \sim \mu$.

Step 4: explicit Radon–Nikodym derivative. Starting from (**??**) and performing the standard change-of-measure computation, for any bounded measurable F,

$$\begin{aligned}\int F(u)\,(T^*_\psi\mu)(du) &= \int F(u+\psi)\,\mu(du) = Z^{-1}\int F(u+\psi)\,e^{-\mathcal{V}(u)}\,\nu(du)\\ &= Z^{-1}\int F(v)\,e^{-\mathcal{V}(v-\psi)}\,(T^*_\psi\nu)(dv),\end{aligned}$$

where we set $v = u+\psi$. Using (**??**) to write $(T^*_\psi\nu)(dv) = \frac{d(T^*_\psi\nu)}{d\nu}(v)\,\nu(dv)$ and then comparing to $\mu(dv) = Z^{-1}e^{-\mathcal{V}(v)}\nu(dv)$ yields

$$(T^*_\psi\mu)(dv) = \left[e^{-\mathcal{V}(v-\psi)}\frac{d(T^*_\psi\nu)}{d\nu}(v)\,e^{\mathcal{V}(v)}\right]\mu(dv).$$

This is exactly (**??**) (rename v as u). Since the exponent is a real-valued random variable and the exponential is strictly positive, the derivative is strictly positive and finite μ-a.s. The corresponding statement for $(T^*_\psi\mu) \ll \mu$ and $\mu \ll (T^*_\psi\mu)$ follows by applying the same computation with $-\psi$ (or equivalently by observing that $T^*_{-\psi}$ is the inverse map). □

An explicit Wick-polynomial expansion (optional)

If one wishes to expand $\mathcal{V}(u) - \mathcal{V}(u-\psi)$, the Wick binomial identities (valid for deterministic ψ) yield

$$:(u-\psi)^2: =: u^2: -2\psi\,u + \psi^2, \qquad :(u-\psi)^4: =: u^4: -4\psi:u^3: +6\psi^2:u^2: -4\psi^3 u + \psi^4.$$

Substituting into (**??**) gives the concrete formula

$$\begin{aligned}\mathcal{V}(u) - \mathcal{V}(u-\psi) = \int_{\mathbb{T}^3}\Big(&\lambda\,\psi\,:u^3: - \frac{3\lambda}{2}\,\psi^2\,:u^2: + \lambda\,\psi^3\,u - \frac{\lambda}{4}\,\psi^4\\ &+ \rho\,\psi\,u - \frac{\rho}{2}\,\psi^2\Big)\,dx,\end{aligned} \tag{5}$$

where the integrals involving u, $:u^2:$, and $:u^3:$ are again interpreted as distribution pairings with smooth test functions. In particular, each term is finite μ-a.s. (indeed in $L^p(\mu)$ for all finite p), so the Radon–Nikodym derivative (**??**) is a strictly positive finite random variable.

Remark 2 (Scope of the statement)**.** *The argument shows more generally that $T_\psi^*\mu \sim \mu$ for every ψ in the Cameron–Martin space $H^1(\mathbb{T}^3)$. For shifts $\psi \notin H^1$, even the free field ν fails to be quasi-invariant (and typically becomes mutually singular to its translate), so one should not expect quasi-invariance of μ outside H^1.*

Conclusion

For every smooth (nonzero) $\psi : \mathbb{T}^3 \to \mathbb{R}$, the Φ_3^4 measure μ on $\mathcal{D}'(\mathbb{T}^3)$ is quasi-invariant under the translation $u \mapsto u + \psi$, and hence μ and $T_\psi^*\mu$ are equivalent.